

Introduction

This work introduces a novel approach for time series representation learning through a contrastive learning framework that leverages temporal structure and feature similarity. The proposed method jointly optimizes the contrastive and MSE objectives, ensuring that the learned representations capture meaningful spatial patterns and temporal information.

Proposed Architecture: DynaCL

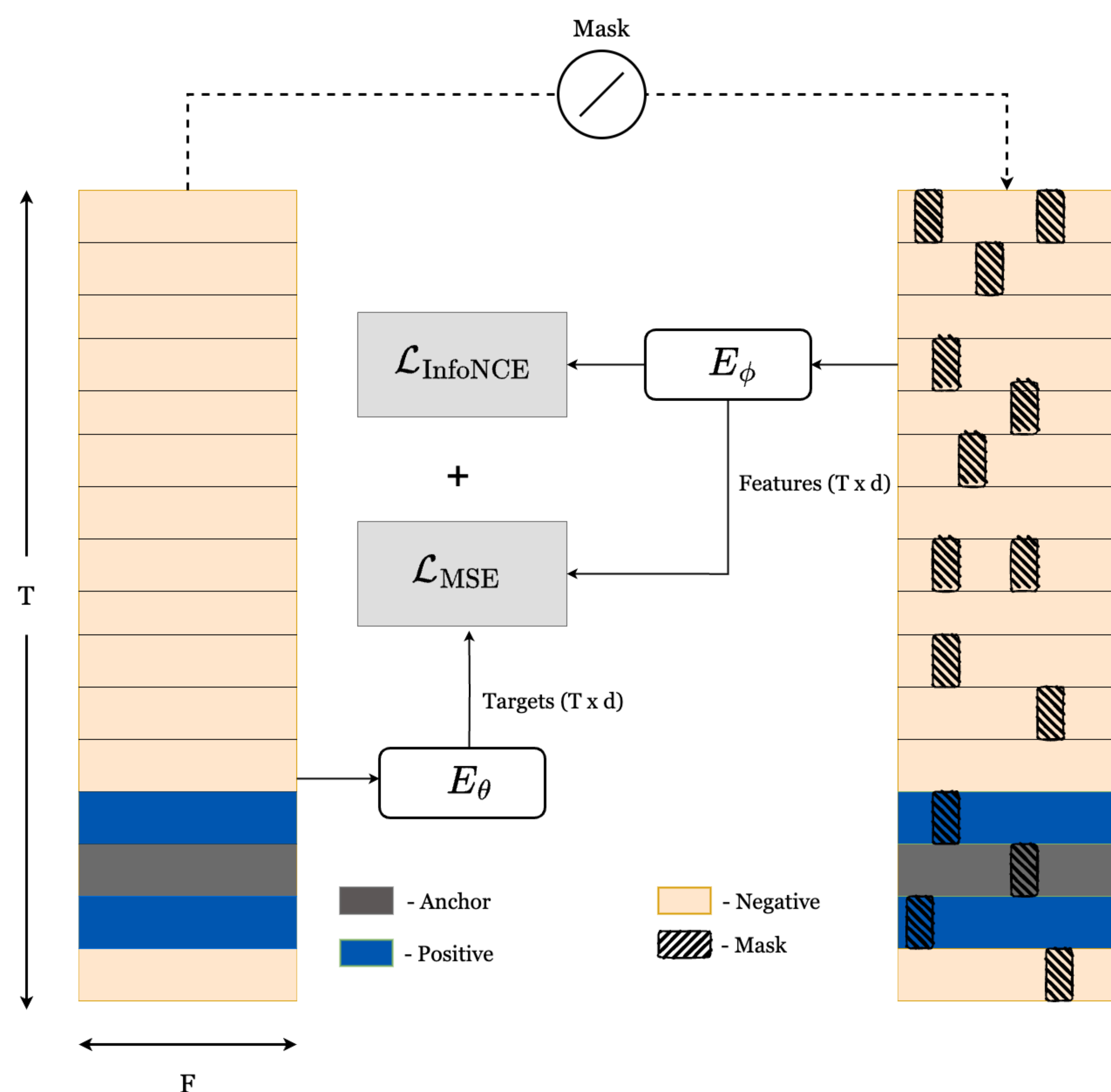


Figure 1: DynaCL: unsupervised time representation learning combining MSE and InfoNCE objectives loss..

Combined objectives

$$\mathcal{L}_{\text{overall}} = \alpha \cdot \mathcal{L}_{\text{MSE}} + \beta \cdot \mathcal{L}_{\text{InfoNCE}} \quad (1)$$

Key Contributions

- Combining the contrastive learning objective and the MSE loss for learning sequential data representation
- Developing multiple adjacent positive example for the infoNCE contrastive learning objective for faster convergence

Results

Table 1: Comparison of clustering metrics (DBI, CHI, SC) with recent state-of-the-art methods on the HAR and EEG datasets

	DBI	HAR CHI	SC	DBI	EEG CHI	SC
CPC	4.71	488.96	-0.02	21.03	13.12	-0.03
TNC	4.91	116.42	-0.19	23.34	26.45	-0.22
InfoTS	4.61	216.64	-0.15	13.98	6.67	-0.14
TS2Vec	4.98	302.07	-0.11	12.34	23.12	-0.09
DynaCL	3.04	829.48	0.09	6.62	147.23	-0.04

Our DynaCL method dynamically adapts to various datasets and achieves impressive results across all three clustering metrics.

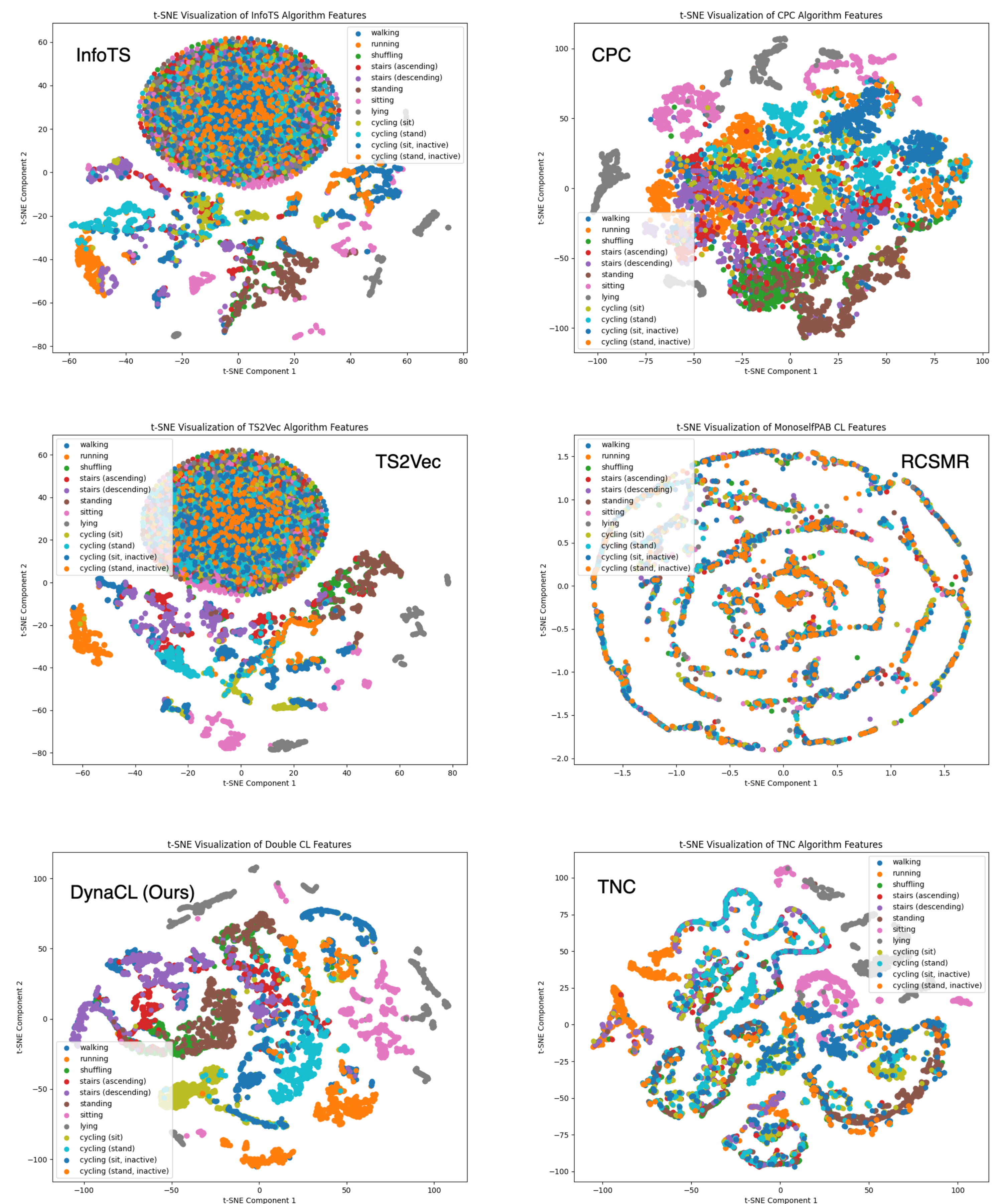


Figure 2: t-SNE visualization of learned feature representations by our DynaCL method and the baselines on the human activity recognition (HAR) dataset

Comparing our features with state-of-the-art baselines in time series representation learning [1], our approach learns a representation space where temporal adjacent activities are placed near one another and two distinct activity in time are placed far apart, thereby producing useful linearly separable feature space.

Conclusion

In this research, we build upon advances in contrastive learning and unsupervised representation learning to learn semantically meaningful embeddings to improve downstream classification performance. Due to the time-contrastive nature of our pretext task, the representation is useful off the shelf. Our findings indicate that our DynaCL loss objective produced features that achieved top scores on 3 clustering metrics.

Acknowledgment

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References

- [1] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu. Ts2vec: Towards universal representation of time series. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 8980–8987, 2022.